Type MS · Explain how temperature or pH changes enzyme activity

Answer 1 of 14 · Question 4 part b · 5 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number				
4	(a)	• bell-shaped line (1) • peak at 37°C ± 3°C (1)		2
	(b)	• increase in temperature (1) • molecules have more (kinetic) energy (1) • more collisions (1) • more enzyme-substrate complexes formed (1) • greater rate of activity (1)		5
	(c)	• hot (1) • because optimum temperature is between 60-75°C (1)		2

Total for this question = 9 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 1 of 14 · Question 4 part b · 5 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 2 of 14 · Question 3 part b.ii · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
3	(a) (i)	A (amino acids); B and D are components of lipids C is a carbohydrate		1
	(ii)	• add Biuret solution to test solution; • mauve/lilac, protein present; • remains blue, protein absent;		3
	(b) (i)	• suitable axes labels; • suitable scale; • correct plots; • suitable curves; • curves labelled;	max 3 for bar chart accept key	5
	(ii)	• denaturation; • loss of shape of active site/enzyme; • substrate no longer fits/decreases rate of activity;		3
	(iii)	any three from • B is much more active/works faster; • over whole temperature range; • higher optimum temperature for B/works fastest at 80°C; • active over a greater range of temperatures;		3

Total 15 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 2 of 14 · Question 3 part b.ii · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 3 of 14 · Question 3 part b · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

number				
3	(a) (i)	water;		1
	(ii)	• allows urea to pass; • prevents other larger molecules from passing;	Allow small molecules go through	2
	(iii)	• breaks down urea; • to produce ammonium ions; • to react with/stimulate sensor;	Allow ammonia Allow idea of being picked up/detected by sensor	3
	(b)	• enzyme activity increases/enzyme works faster/higher rate of reaction; • at higher temperatures; • more ammonium ions produced; • higher reading;	Ignore details related to enzyme denaturing / collision theory/kinetic energy ORA for lower temperature	4
	(c)	any two from • can be reused; • products not contaminated; • enzymes more stable; • cost effective justified;	Allow can withstand temperature/pH changes	2

Total 12 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 3 of 14 · Question 3 part b · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 4 of 14 · Question 4 part b.iii · 4 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

4 (a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
(ii)	lipase		1
(iii)	fatty acids; glycerol;		2
(b) (i)	Any value between 8.6 and 8.8;		1
(ii)	independent = time/ temperature; dependent = pH;		2
(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Explain how temperature or pH changes enzyme activity

Answer 4 of 14 · Question 4 part b.iii · 4 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 5 of 14 · Question 5 part b · 4 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

5 (a) (i)	any five from • place substrate into a test/boiling tube/beakers; • place test tube in a (hot) water bath; • add enzyme to substrate; • measure time taken for reaction to occur; • use same volume/concentration of enzyme/substrate; • repeat for other temperatures;	Allow a temperature value Allow example of a reaction e.g. colour change, amount of glucose produced reject amount	(5)
(ii)	temperature;		(1)
(b)	• less kinetic energy; • less/fewer collisions (with substrate); • less enzyme-substrate complexes formed/less binding; • lower enzyme activity/rate of reaction;		(4)

Total for question 5 = 10 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 6 of 14 · Question 8 part c · 5 marks

2024 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

8 (a)	any three from • released from pancreas; • when glucose levels rise; • causes/stimulates conversion of glucose to glycogen; • in liver/muscles;		(3)
(b) (i)	• vasodilation; • more blood flows at/near the surface of the skin; • heat (from blood) radiated (away from body)/increased heat loss; • hairs lie flat; • less air trapped; • (more) heat radiated;		(3)
(ii)	vasoconstriction/blood vessels in the skin narrowed shivering/increased metabolic rate/hairs stand on end;		(1)
(iii)	• hypothalamus detects rise/fall/change in body temperature; • (named) mechanism reverses the change/brings back to set point/normal;	e.g. vasodilation, increase in sweating	(2)
(c)	• enzymes lose kinetic energy; • enzyme activity decreases/works slowly; • less enzyme substrate complexes formed/ less collisions between enzyme and substrate/binding with substrate; • rate of chemical reactions/metabolism slow down; • reference to symptom of hypothermia e.g. death/weak pulse/slower heart/breathing rate;		(5)

Total for Question 8 = 14 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 7 of 14 · Question 7 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 7 of 14 · Question 7 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Explain how temperature or pH changes enzyme activity

Answer 8 of 14 · Question 3 part c · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

3 (a)	• control of/maintenance of • constant internal environment; • despite changes in external environment; • ref negative feedback;	ignore specific examples	4
(b)	box 1 hypothalamus; detects rise/increase in blood temperature; box 2 arterioles dilate/vasodilation; box 3 sweat evaporates; box 4 (more) heat loss (from skin); by radiation;	ignore thermoregulatory centre	6
(c)	• enzymes; • denatured; • fewer/no enzyme substrate complexes formed/substrate can't bind to active site/reduced enzyme activity/reaction; • reduced metabolic activity/rate of reactions in body reduced;		4
			14 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 8 of 14 · Question 3 part c · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 9 of 14 · Question 2 part c · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 9 of 14 · Question 2 part c · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 10 of 14 · Question 1 part b.iii · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
1	(a) (i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);		3 marks
	(ii)	• no repeats/only done once;		1 mark
	(iii)	• results based on judgement/subjective/qualitative;		1 mark
	(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;		3 marks
	(b) (i)	untreated/room temperature;		1 mark
	(ii)	control;		1 mark
	(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;		4 marks

Total 14 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 10 of 14 · Question 1 part b.iii · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 11 of 14 · Question 6 part b · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 11 of 14 · Question 6 part b · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 12 of 14 · Question 2 part b.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 13 of 14 · Question 4 part a.v · 3 marks

2024 JUN · Paper 01 (regional R) · find part a.v in this mark scheme extract

Number				
4 (a)	(i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
(b)		• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Explain how temperature or pH changes enzyme activity

Answer 14 of 14 · Question 4 part b.ii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%;	MP1- accept Vmax	2
			• all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	not enough/no more substrate/protein to bind to active sites= 2 marks	
					Total = 15

Type MS · Explain how temperature or pH changes enzyme activity

Answer 14 of 14 · Question 4 part b.ii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks